

A Dimension-Dependent Subcase of Ostrovskii's Projection-Factorization Problem

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Status

This packet records a likely valid strengthened partial result for Problem 3 in M. I. Ostrovskii, *Compositions of projections in Banach spaces and relations between approximation properties*, arXiv:0811.1763. The result does not solve the full problem, which asks for a constant independent of all dimensions. It proves the problem with a constant depending only on $d = \dim X_1$, namely $C_d = 1 + 2\sqrt{d}$. In particular the line case has $C_1 = 3$.

Source Problem

Ostrovskii asks the following problem on page 7 of the source paper.

Remark. By [8, Theorem 5] the sufficient enlargement $A = \lambda(\ell_2^k)B_2^k + \lambda(\ell_2^{n-k})B_2^{n-k}$ is minimal. Hence $\text{cl}(P_2(B_{L_\infty(\mu)})) = A$ and $\|P_2\| = ((\lambda(\ell_2^k))^2 + (\lambda(\ell_2^{n-k}))^2)^{1/2}$.

Is it always like this? More precisely

Problem 3 Does there exist a universal constant $C \in [1, \infty)$ such that for each triple $X_1 \subset X_2 \subset X_3$ of Banach spaces, with X_1 and X_2 finite dimensional, there exist projections $P_1 : X_2 \rightarrow X_1$ and $P_2 : X_3 \rightarrow X_2$, such that $\|P_2\| \leq C\lambda(X_2, X_3)$ and $\|P_1P_2\| = \lambda(X_1, X_3)$?

Another version of this problem (which will be particularly interesting if Problem 3 has a negative answer):

In the paper's notation, $\lambda(Y, X)$ denotes the relative projection constant of a finite-dimensional subspace Y in a normed space X .

Idea of the Proof

Start with a minimal projection $P : X_3 \rightarrow X_1$ and a minimal projection $R : X_3 \rightarrow X_2$. The product $(P|_{X_2})R$ usually does not equal P . The defect is the finite-rank map $P - (P|_{X_2})R$, which vanishes on X_2 and has range in $X_1 \subset X_2$. Adding this defect to R keeps the map fixed on X_2 , hence keeps it a projection onto X_2 , while forcing the composed projection onto X_1 to be the chosen minimal projection P . The price is controlled by $\lambda(X_1, X_3)$. The classical Kadets-Snobar theorem says that every d -dimensional subspace of a Banach space is $\sqrt{d}$ -complemented [2]. This gives a constant depending only on $d = \dim X_1$.

Theorems

Theorem 1 (Correction theorem). *Let $X_1 \subset X_2 \subset X_3$ be Banach spaces with X_1 and X_2 finite-dimensional. Then there exist projections $P_1 : X_2 \rightarrow X_1$ and $P_2 : X_3 \rightarrow X_2$ such that*

$$\|P_2\| \leq (1 + 2\lambda(X_1, X_3))\lambda(X_2, X_3) \quad \text{and} \quad \|P_1 P_2\| = \lambda(X_1, X_3).$$

Proof. Because X_1 and X_2 are finite-dimensional, the relative projection constants are attained. Let $P : X_3 \rightarrow X_1$ and $R : X_3 \rightarrow X_2$ be projections with

$$\|P\| = \lambda(X_1, X_3), \quad \|R\| = \lambda(X_2, X_3).$$

Define

$$P_1 = P|_{X_2} : X_2 \rightarrow X_1$$

and

$$P_2 x = Rx + Px - P(Rx), \quad x \in X_3.$$

The map P_1 is a projection from X_2 onto X_1 , since P is the identity on X_1 . The map P_2 has range contained in X_2 , because both Rx and $Px - P(Rx)$ lie in X_2 . If $x \in X_2$, then $Rx = x$, and hence $P_2 x = x + Px - Px = x$. Thus P_2 is a projection from X_3 onto X_2 .

For every $x \in X_3$,

$$P_1 P_2 x = P(Rx) + P(Px) - P(P(Rx)) = Px.$$

Thus $P_1 P_2$ is the minimal projection P onto X_1 , and

$$\|P_1 P_2\| = \lambda(X_1, X_3).$$

For $\|x\| \leq 1$,

$$\|P_2 x\| \leq \|Rx\| + \|Px\| + \|P(Rx)\| \leq \lambda(X_2, X_3) + \lambda(X_1, X_3) + \lambda(X_1, X_3)\lambda(X_2, X_3).$$

Since every projection onto a nonzero subspace has norm at least one, $\lambda(X_2, X_3) \geq 1$. Therefore

$$\|P_2\| \leq (1 + 2\lambda(X_1, X_3))\lambda(X_2, X_3).$$

This proves the theorem. □

Theorem 2 (Fixed-dimensional corollary). *If $\dim X_1 = d$, then Ostrovskii's Problem 3 has an affirmative answer for the triple $X_1 \subset X_2 \subset X_3$ with*

$$C_d = 1 + 2\sqrt{d}.$$

In particular, if $\dim X_1 = 1$, the problem has an affirmative answer with $C_1 = 3$.

Proof. By the Kadets-Snobar theorem, every d -dimensional subspace of a Banach space is $\sqrt{d}$ -complemented. Hence $\lambda(X_1, X_3) \leq \sqrt{d}$, and the preceding theorem gives

$$\|P_2\| \leq (1 + 2\sqrt{d})\lambda(X_2, X_3)$$

while preserving $\|P_1 P_2\| = \lambda(X_1, X_3)$. □

Verification Notes

The correction theorem uses Hahn-Banach-level finite-rank projection algebra and the standard attainment of relative projection constants for finite-dimensional ranges. The latter can be proved by writing projections onto the finite-dimensional target through a fixed finite basis and using weak-star compactness of bounded sets in X_3^* to pass to a norm-minimizing limit. The fixed-dimensional corollary uses the classical Kadets-Snobar projection theorem.

No computational verification is needed. A bounded novelty check searched the run indexes for arXiv:0811.1763 and projection-factorization keywords, searched the local parsed corpus for the problem phrases, and searched the web for exact phrases from Problem 3. No prior packet or obvious published resolution of this dimension-dependent correction theorem was found. The novelty confidence is modest, because the argument is elementary and the full problem is much stronger.

Limitations

The result is not a full solution of Problem 3. The constant obtained here depends on $\sqrt{d}$, where $d = \dim X_1$ (or more sharply on $\lambda(X_1, X_3)$). The original problem asks for a single universal constant independent of d , X_1 , X_2 , and X_3 .

Human Review Recommendation

Review the correction formula for P_2 and the norm estimate. The most important point is that the correction $P - (P|_{X_2})R$ vanishes on X_2 , which makes P_2 a genuine projection onto X_2 while forcing P_1P_2 to equal the chosen minimal projection onto X_1 .

References

- [1] M. I. Ostrovskii, *Compositions of projections in Banach spaces and relations between approximation properties*, arXiv:0811.1763, 2008.
- [2] M. I. Kadets and M. G. Snobar, *Some functionals over a compact Minkovskii space*, Mathematical Notes **10** (1971), 694–696.